SunMint Program — Consolidated Progress Report (v8)

Prepared by: Sophia Truesight · **Date:** 2026-08-25 · **Status:** Execution has started

1. The model has evolved — phone-first MRV

- **Native Android/iOS app SHIPPED** (2026-08-23, Capacitor 8, offline-first, RSA-signed, PT/EN) — the farmer's phone is now the carbon-measurement device
- **Primary measurement layer:** farmer smartphone tree-growth monitoring (DBH/height/species via calibration-card photo, per ACORN/CommuniTree/TREEO operating model) — TREEO 94–95% DBH accuracy ($R^2 \geq 0.95$)
- **Free satellite tier (support):** Copernicus Data Space Ecosystem (Sentinel-2, 10 m, 5-day, free commercial use) · Microsoft Planetary Computer · USGS EarthExplorer (Landsat 1972→, public domain) — for VM0047 non-forest ≥10-yr baseline + annual leakage context, at $0 software cost
- **Drones (PODream):** deferred to post-pilot calibration sampling on a minority of plots
- **NOR Space:** optional premium AI analysis — only if the VVB demands it

2. Certification decision — Plan Vivo FIRST

- **Plan Vivo PV Climate is CCP-Eligible under ICVCM** — the same benchmark Verra's top methodologies hold
- **High-rated PVCs traded at median > €30/t in 2025** — above the broad VCU market (buyers pay premium for community co-benefits)
- **ACORN and CommuniTree — our strongest precedents — certify under Plan Vivo**, not Verra
- **Two-registry portfolio:** Plan Vivo for smallholder plots (faster, cheaper, community-aligned, ≥60% revenue to communities) · Verra VM0047 + CCB only at scale or when an offtake buyer demands VCS liquidity
- Precedent: Andean Cacao (Colombia) — first large-scale cacao ARR verified (VM0047 + CCB, 56,000+ VCUs first issuance, Terra Global)

3. The accounting methodology (PM002 — found & verified)

From Plan Vivo PM002 v1.0 (ACORN's certified smallholder agroforestry methodology, active 29/09/2025):

- **DBH + species → allometric equation → AGB** (per-tree, implementable in the app)
- **Eq. 1:** $\Delta BGB\_p = \Delta AGB\_p \times R$ (root:shoot; IPCC default $R = 0.32$)
- **Eqs. 6.1/6.2:** $PVCs = ((\Delta AGB\_p + \Delta BGB\_p) \times 0.47) \times 44/12 \times (1 - A\_pre)(1 - A\_unc)(1 - LD)(1 - AR)(1 - RB) - (E\_proj - E\_base)$
- 0.47 = carbon fraction · 44/12 = CO■ conversion
- A_pre = pre-issuance buffer · A_unc = uncertainty · LD = leakage deduction
- **AR = 10% achievement reserve · RB = 20% risk buffer**

4. RESOURCE GAPS (what's needed to execute)

#	Gap	Cost/Effort	Filled by
1	PDD/PM002 writer	$8–15k	Grant (below) + **cacao sales**
2	VVB engagement (AENOR/Earthood)	$10–20k	Grant + **cacao sales** + credit pre-sale
3	Baseline proof (non-forest ≥10yr)	FREE satellite archive + 2–4 wk GIS labor	Copernicus/USGS/Planetary Computer
4	GIS plot boundaries	2–4 wk GIS labor	Grant (Gitcoin/CFC)
5	monitor_tree_growth module	2–3 wks dev	In-house (DAO dev)
6	Allometric equations (5 species)	$5–10k	Fundo Vale / iNovaland
7	CAR tenure + FPIC records	1–2 mo, co-op coordination	Cocoa Horizons / Cocoa Life
8	Stage 0 capital	**$5–10k** (dropped — NOR removed)	DAO injection + Gitcoin

5. FUNDING — existing grants + the cacao-sales flywheel

Grants (no external investors needed):

- **Cocoa-industry funds:** Cocoa Horizons (Barry Callebaut) · Cocoa Life (Mondelēz, $1B → 2030) · Cargill Cocoa Promise · Fundo Vale (Belterra/Caapora cacao agroforestry incubators)
- **Flagship:** GCF + IICA cacao program — 12,500 ha, 5.18 Mt CO_2e, 3,000 producers — SunMint's model at scale
- **Regional:** iNovaland South Bahia restoration call (R$8.8M) · Fundo Floresta Atlântica
- **ReFi/fast:** Gitcoin climate rounds (US$200k+ pools) · CFC grants (cocoa focus) · Forest Conservation Fund (quarterly) · regen.fund
- **Blockers are paperwork + sequencing, not money.**

The cacao-sales self-funding flywheel (new):

- A share of **cacao/chocolate sales is reserved to fund the carbon-certification pipeline** (PDD development, VVB validation, monitoring) **alongside the share reserved for planting trees**
- **The more chocolate sold → the more credits can be issued** — product revenue closes the loop between sales and carbon infrastructure
- Grants cover Stage 0–1; **cacao sales build recurring self-funding** for ongoing certification costs

6. MILESTONES — executed vs planned

Executed ✓

- **M0** Native Android/iOS app shipped (2026-08-23)
- **M1** First tree linked to a sold QR code (2026-08-22)
- **M2** PDD methodology corrected to VM0047/PM002 (PRs #294, #307, #308, #309, #310, #311)

Planned

- **M3** Stage 0 capital + grant applications — Q3 2026
- **M4** PDD draft + free-satellite baseline — Q4 2026
- **M5** monitor_tree_growth module — Q4 2026
- **M6** VVB engagement + Plan Vivo validation — Q1 2027
- **M7** Pilot planting 20–50 ha — 2027
- **M8** Terra RFP / offtake + scale — 2027+

7. EXECUTION HAS STARTED — the first funded tree

- **QR code:** FOUNDERHAUS_BOUGAINVILLEA_20260821_1
- **Currency:** SunMint Tree Planting Pledge — QR Code · **Status:** ASSIGNED_TO_TREE
- **Owner:** paloma@founderhaus.club · **Manager:** Gary Teh
- **Ledger:** https://truesight.me/sunmint/main
- **Linked:** 2026-08-22 (first real non-test tree–QR link)
- **Scan URL:** https://edgar.truesight.me/agroverse/qr-code-check?qr_code=FOUNDERHAUS_BOUGAINVILLEA_20260821_1

The program is real. The first tree is planted and on-ledger. The phone-MRV model is live. Now we fund, validate, and scale.

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